Keywords: multimodal learning • tensor decomposition • synergy • contrastive learning

HRIL: Learning Multimodal Synergy via Higher-Order Tensor Modeling

Tucker Decomposition of a Cross-Moment Tensor Captures the Emergent Information That Pairwise Contrastive Objectives Miss

By Qun Dai, Zhao Kang et al.

arXiv:2610.12393 • NeurIPS 2026 • October 2026

Multimodal contrastive learning fails to capture *synergistic* information—the signal that is only present in the joint combination of three or more modalities. HRIL introduces a cross-moment tensor over modality embeddings and factorises it with Tucker decomposition inside a contrastive objective, with a regulariser that prevents the core tensor from collapsing to pairwise terms.

AT A GLANCE

Problem: Pairwise contrastive losses miss synergistic information that only exists in the joint combination of three or more modalities.

Why it matters: Many real-world multimodal tasks rely on cross-modal synergy; ignoring it leaves performance on the table.

Method: Outer-product cross-moment tensor over modality embeddings, compressed by Tucker decomposition, with a diagonal-suppressing regulariser.

Result: Consistent gains over multimodal contrastive baselines on synergy-dominated benchmarks; accepted to NeurIPS 2026.

Evidence: Ablation confirms the synergy regulariser is necessary; Tucker rank controls the expressiveness–cost trade-off.

The Big Picture

The information-theoretic decomposition of multimodal data identifies three components: *redundant* (shared across modalities), *unique* (present in one modality only), and *synergistic* (present only in the joint combination). Standard contrastive objectives such as InfoNCE maximise pairwise mutual information, capturing redundant and unique information but systematically missing synergy.

HRIL (Higher-order Representation Interaction Learning) addresses this gap by constructing a tensor representation that explicitly encodes interactions of all orders up to M for M modalities, then compressing it efficiently via Tucker decomposition. The result is a contrastive framework that is provably sensitive to synergistic effects.

Why Existing Approaches Fall Short

Pairwise contrastive losses (InfoNCE, SimCLR) consider only two modalities at a time, so terms requiring all M modalities simultaneously are never optimised.

Late fusion combines independent unimodal embeddings at inference time, encoding no interactions before the final decision.

Bilinear tensor fusion captures pairwise interactions and scales quadratically with the number of modalities, missing higher-order terms and becoming intractable for large M.

Core Mechanism

HRIL forms the outer product of modality embeddings:

$$\mathcal{T}^{(i)} = \bigotimes_{m=1}^{M} \mathbf{z}_m^{(i)}, \quad \mathcal{T}^{(i)} \in \mathbb{R}^{d_1 \times \cdots \times d_M}$$

and compresses it via Tucker decomposition into a core tensor $\mathcal{G} \in \mathbb{R}^{r_1 \times \cdots \times r_M}$ with factor matrices $\mathbf{U}_m \in \mathbb{R}^{d_m \times r_m}$. A regulariser penalises the diagonal entries of $\mathcal{G}$ (which correspond to unimodal and pairwise terms), forcing the model to allocate capacity to genuinely multi-way interactions.

Technical Formulation

Let $f_m : \mathcal{X}_m \to \mathbb{R}^{d_m}$ be the encoder for modality m, with output $\mathbf{z}_m^{(i)} = f_m(x_m^{(i)})$.

Tucker factorisation approximates the cross-moment tensor:

$$\hat{\mathcal{T}} = \mathcal{G} \times_1 \mathbf{U}_1 \cdots \times_M \mathbf{U}_M, \quad \|\mathcal{T} - \hat{\mathcal{T}}\|_F \leq \epsilon$$

The synergy regulariser penalises diagonal concentration:

$$\mathcal{L}_{\text{syn}} = \sum_r \mathcal{G}^2_{r,r,\ldots,r}$$

The total objective balances contrastive alignment and synergy:

$$\mathcal{L} = \mathcal{L}_{\text{InfoNCE}} + \lambda \mathcal{L}_{\text{syn}}$$

where $\lambda > 0$ is a hyperparameter.

Learning or Inference Procedure

Training:

1. Encode each modality: $\mathbf{z}_m^{(i)} = f_m(x_m^{(i)})$.
2. Form the outer-product tensor $\mathcal{T}^{(i)}$.

3. Project through Tucker factors to obtain the core $\mathcal{G}^{(i)}$.
4. Evaluate the contrastive loss over positive/negative pairs.
5. Add $\lambda\mathcal{L}_{\text{syn}}$ and backpropagate through Tucker factors and encoders jointly.

Inference: The Tucker-compressed representation provides a fused embedding for downstream classification or retrieval.

What the Guarantee Says

Tucker decomposition provides a controlled approximation: the Frobenius reconstruction error is bounded by the residual from truncating the Tucker rank. The synergy regulariser provably prevents the core from degenerating to a diagonal form (which recovers bilinear fusion).

Experimental Findings

- HRIL outperforms multimodal contrastive baselines on synergy-dominated benchmarks, with gains largest when no pair of modalities alone captures the label.
- Performance scales with Tucker rank $(r_1, \ldots, r_M)$ up to a saturation point, providing a tuneable compute–accuracy trade-off.
- Gains transfer across different encoder architectures and modality combinations.

Ablations and Interpretation

Without synergy regulariser: The core concentrates on diagonals, recovering bilinear-fusion behaviour and losing most of the HRIL advantage.

Tucker rank: Lowering the rank reduces cost with minor accuracy loss; higher ranks plateau beyond a problem-dependent threshold.

Number of modalities: The benefit over pairwise methods grows with M, consistent with the theoretical argument that synergy is a higher-order effect.

Reference

Qun Dai, Zhao Kang et al. **HRIL: Learning Multimodal Synergy via Higher-Order Tensor Modeling**. NeurIPS 2026. arXiv:2610.12393, October 2026.
https://arxiv.org/abs/2610.12393

Keywords: mixture of experts • interpretability • sparse autoencoders • routing

RouterInterp: Understanding Superposed Specialisation in Mixture of Experts Routing

Experts Do Not Own Domains—They Activate on Disjoint Unions of Fine-Grained Features, and Sparse Autoencoder Probes Explain Why

By Ilya Lasy, Nora Yinuo Cai & Kola Ayonrinde

arXiv:2610.11775 • ICML 2026 • October 2026

MoE routing does not follow the single-domain assumption: each expert activates on a *disjoint union* of fine-grained features—the Superposed Specialisation Hypothesis. RouterInterp identifies these features via Sparse Autoencoder probes and composes them into natural-language routing explanations, achieving 65% higher routing-detection accuracy than token-statistics baselines on gpt-oss-20b.

AT A GLANCE

Problem: It is unknown why specific tokens are routed to specific experts in MoE language models.

Why it matters: Understanding routing is essential for auditing, debugging, and improving MoE architectures.

Method: Sparse linear probes over SAE features predict routing; top-feature descriptions are composed into expert explanations.

Result: ≈65% higher routing-detection accuracy on gpt-oss-20b vs. token-statistics baselines.

Evidence: Causal interventions confirm the SAE features causally influence routing; manual validation confirms explanation quality.

The Big Picture

Mixture-of-Experts (MoE) models achieve state-of-the-art scaling efficiency by routing each token through a small subset of expert feed-forward networks. Despite widespread deployment, the routing mechanism remains a black box: knowing *why* a token is routed to a particular expert is important for diagnosing failure modes, ensuring balanced expert use, and building trust in deployed systems.

The dominant assumption—"the code expert," "the French expert," etc.—is a convenience fiction. RouterInterp provides empirical and causal evidence for a richer picture: the **Superposed Specialisation Hypothesis (SSH)**.

Why Existing Approaches Fall Short

Token-statistics methods compute the frequency distribution of token types per expert and label each with the most common category. These methods are coarse, conflate co-occurrences with specialisation, and show low prediction accuracy when routing is polysemous.

Probing classifiers on raw activations predict routing but yield no human-readable explanation.

Manual inspection is intractable across the hundreds of experts and billions of parameters in production MoE models.

Core Mechanism

Superposed Specialisation Hypothesis: Each expert e activates not on one broad domain, but on a disjoint union of fine-grained linguistic or semantic features—mirroring the superposition phenomenon known to occur in individual neurons.

RouterInterp operationalises this via Sparse Autoencoders (SAEs):

1. Derive monosemantic SAE features from the residual stream.
2. Fit L_1-penalised probes to predict routing from SAE activations.
3. Compose the top-predictive features' descriptions into a unified expert explanation.

Technical Formulation

Let $\mathbf{h}_t \in \mathbb{R}^d$ be the residual stream at token t. A Sparse Autoencoder maps it to a sparse feature vector:

$$\mathbf{f}_t = \mathrm{ReLU}(\mathbf{W}_{\mathrm{enc}}\,\mathbf{h}_t + \mathbf{b}_{\mathrm{enc}}), \quad \mathbf{f}_t \in \mathbb{R}^F$$

The routing indicator for expert e is:

$$r_{t,e} = \mathbf{1}[\text{expert } e \text{ selected for token } t]$$

RouterInterp solves the L_1-penalised logistic regression:

$$\hat{r}_{t,e} = \sigma(\boldsymbol{\beta}_e^\top \mathbf{f}_t), \quad \boldsymbol{\beta}_e = \arg\min_{\boldsymbol{\beta}} \mathcal{L}_{\mathrm{CE}}(r, \hat{r}) + \mu\|\boldsymbol{\beta}\|_1$$

Non-zero entries $\{k : \beta_{e,k} \neq 0\}$ identify the SAE features predictive of routing to expert e. Their descriptions $\{s_k\}$ are composed into a natural-language explanation via a language model.

Learning or Inference Procedure

1. Train a SAE on the frozen MoE's residual stream to obtain a dictionary of monosemantic features.

2. Collect routing labels $r_{t,e}$ over a large corpus.
3. Fit sparse probes $\boldsymbol{\beta}_e$ for each expert over SAE feature activations.
4. Extract the top-k predictive features per expert.
5. Pass the feature descriptions to a language model to compose a coherent expert summary.

What the Guarantee Says

RouterInterp provides an empirical claim validated by causal intervention: activating a high-weight SAE feature at a token that would otherwise not route to expert e measurably increases routing probability to e, confirming causality beyond mere correlation.

Experimental Findings

- On **gpt-oss-20b**, RouterInterp achieves **≈65% higher routing-detection accuracy** than token-statistics baselines.
- Manual inspection confirms that expert descriptions are specific, coherent, and capture non-obvious expert roles.
- The SSH is validated: the top SAE features per expert span semantically disjoint categories (e.g., medical terminology *and* certain syntactic patterns in Spanish for the same expert).
- Results are consistent across all expert layers in gpt-oss-20b.

Ablations and Interpretation

Raw activations vs. SAE features: Probing on raw residual stream activations degrades routing prediction accuracy significantly and yields incoherent explanations.

Top-k features: Including more features beyond the natural coefficient elbow adds noise without improving routing accuracy.

Causal intervention: Artificially activating a predictive SAE feature causally increases routing probability to the predicted expert.

Across layers: Early layers show routing correlated with syntactic SAE features; later layers show richer semantic specialisation.

Reference

Ilya Lasy, Nora Yinuo Cai, Kola Ayonrinde. **RouterInterp: Understanding Superposed Specialisation in Mixture of Experts Routing**. ICML 2026. arXiv:2610.11775, October 2026.
https://arxiv.org/abs/2610.11775

Keywords: sequential Monte Carlo • particle filters • differentiable resampling • variational inference

Differentiable Systematic Resampling for Variational Sequential Monte Carlo

A Sigmoid Softening of the CDF Ancestor Assignment Recovers Systematic Resampling's Variance Advantage While Allowing Full Gradient Flow

By Fredrik Cumlin & Saikat Chatterjee

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Systematic resampling is the most variance-efficient particle-filter resampling scheme, but its step-function CDF blocks gradient flow. Differentiable Systematic Resampling (DSR) softens the CDF-level ancestor assignment with sigmoid functions, sorts particles via a Morton space-filling curve to preserve the banded ancestor structure, and proves that the bias vanishes exponentially in the inverse temperature—without iterative solvers.

AT A GLANCE

Problem: Systematic resampling blocks gradient flow through variational SMC due to its step-function CDF.

Why it matters: VSMC is a powerful generative modelling tool; its ELBO cannot be fully optimised without differentiable resampling.

Method: Sigmoid CDF softening with Morton-curve particle sorting; bias proven to decay exponentially in $1/\tau$.

Result: Comparable or better filtering and dynamics-learning performance vs. OT-based methods, 3–5× faster per step.

Evidence: Lorenz-96 stochastic dynamics, IAM handwriting dataset; ablation confirms Morton sorting is necessary.

The Big Picture

Sequential Monte Carlo (SMC) approximates temporal posterior distributions with weighted particles $\{(w_t^k, x_t^k)\}_{k=1}^K$. Resampling—selecting ancestors for the next particle generation—prevents degeneracy but introduces a discontinuity that blocks gradients. Variational SMC (VSMC) maximises an ELBO over proposal and model parameters via backpropagation through the full particle trajectory, so differentiable resampling is a key missing ingredient.

Systematic resampling is preferred in practice because it has $O(K^{-2})$ variance in the sample count (vs. $O(K^{-1})$ for multinomial) and is $O(K)$ to compute. DSR makes it differentiable without sacrificing these advantages.

Why Existing Approaches Fall Short

Multinomial resampling (Gumbel-softmax variants) is differentiable but has higher variance; the systematic resampling advantage is lost.

OT-based differentiable resampling preserves the systematic resampling's CDF-ordered structure but requires an iterative Sinkhorn solver at every resampling step, adding substantial overhead.

Stop-gradient resampling avoids differentiating through resampling but biases the ELBO gradient estimator systematically.

Core Mechanism

Standard systematic resampling draws $U \sim \text{Uniform}(0, 1/K)$ and assigns ancestor $a^j = \min\{i : W_{\le i} \ge (j-1)/K + U\}$, where $W_{\le i} = \sum_{l \le i} w^l$.

DSR replaces the step function with a sigmoid:

$$\tilde{A}_{ij} = \sigma\left(\frac{W_{\le i} - (j-1)/K - U}{\tau}\right) - \sigma\left(\frac{W_{\le i-1} - (j-1)/K - U}{\tau}\right)$$

The soft-resampled particle is $\tilde{x}^j = \sum_i \tilde{A}_{ij}\, x^i$.

A Morton (Z-order) space-filling curve sorts particles before resampling so that spatially proximate particles are adjacent in CDF order, preserving the *banded* ancestor matrix property that gives systematic resampling its variance efficiency.

Technical Formulation

Let $\{w^k\}_{k=1}^K$ be normalised weights with prefix sums $W_{\le k} = \sum_{j \le k} w^j$.

The soft ancestor matrix is:

$$\tilde{A}_{ij}(\tau) = \sigma\left(\tfrac{W_{\le i} - \frac{j-1}{K} - U}{\tau}\right) - \sigma\left(\tfrac{W_{\le i-1} - \frac{j-1}{K} - U}{\tau}\right)$$

Bias bound: There exist constants $C, c > 0$ such that

$$\|\mathbb{E}[\tilde{A}(\tau)] - \mathbb{E}[A]\|_\infty \le C e^{-c/\tau}$$

This exponential rate is proved from the pointwise approximation error of the sigmoid to the Heaviside step function.

Learning or Inference Procedure

Variational SMC with DSR:

1. Propagate: $x_t^k \sim q_\phi(x_t \mid x_{t-1}^k, y_t)$.
2. Weight: $w_t^k \propto p_\theta(y_t|x_t^k)\, p_\theta(x_t^k|x_{t-1}^k) / q_\phi(x_t^k|x_{t-1}^k, y_t)$.

3. Sort particles via Morton curve.
4. Apply DSR to obtain $\{\tilde{x}_t^k\}$.
5. Accumulate ELBO: $\mathcal{L} = \mathbb{E}[\log \frac{1}{K} \sum_k w_T^k]$.
6. Backpropagate through soft ancestor assignments to update ϕ, θ.

What the Guarantee Says

The bias bound $O(e^{-c/\tau})$ is stronger than the $O(\tau)$ or polynomial rates of OT-based alternatives. At temperature $\tau = 0.01$, the bias is essentially zero for practical purposes, while gradients remain well-defined through the sigmoid.

Experimental Findings

- On **Lorenz-96 stochastic dynamics**, DSR matches OT-based differentiable resampling on filtering NLL with **3–5× lower** per-step wall-clock time.
- On **IAM handwriting generation**, DSR improves log-likelihood over both multinomial and OT-based resampling.
- The absence of iterative solvers is confirmed as the primary source of the speed advantage.

Ablations and Interpretation

Without Morton sorting: The soft ancestor matrix loses its banded structure; gradient variance increases and ELBO estimates degrade.

Temperature sensitivity: An optimal τ^* balances gradient noise (low τ) vs. bias (high τ); a short sweep identifies it.

Number of particles: The advantage over multinomial resampling grows with K, consistent with the $O(K^{-2})$ vs. $O(K^{-1})$ variance rates.

Stop-gradient comparison: Stop-gradient introduces a systematic ELBO bias that DSR eliminates.

Reference

Fredrik Cumlin, Saikat Chatterjee. **Differentiable Systematic Resampling for Variational Sequential Monte Carlo**. NeurIPS 2026. arXiv:2610.12094, October 2026.
https://arxiv.org/abs/2610.12094

Keywords: Bayesian neural networks • confidence sequences • adaptive inference • uncertainty quantification

When Has a Bayesian Neural Network Sampled Enough? Adaptive Inference Time with Statistical Guarantees

Anytime-Valid Confidence Sequences Terminate Monte Carlo Sampling Per-Input When the Decision Is Statistically Resolved

By Fabian Denoodt & Sibylle Hess

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A fixed Monte Carlo sample budget in Bayesian neural network inference over-samples easy inputs and may under-sample hard ones, with no formal control on decision reliability. This paper terminates sampling per-input using *confidence sequences*—anytime-valid probability bounds—as soon as the relevant decision is statistically resolved, concentrating compute on ambiguous cases and providing a formal per-prediction guarantee.

AT A GLANCE

Problem: Fixed-budget BNN inference leaves finite-sample prediction error uncontrolled and wastes compute on easy inputs.

Why it matters: Reliable per-input guarantees are critical for high-stakes BNN deployment.

Method: Anytime-valid KL-mixture confidence sequences for argmax, distributional, and threshold BNN decisions.

Result: Adaptive sampling reduces latency vs. matched-reliability fixed budgets while providing a formal guarantee per prediction.

Evidence: Image classification benchmarks; comparison with fixed-S and heuristic early stopping.

The Big Picture

Bayesian neural networks approximate the posterior predictive distribution by averaging over S Monte Carlo weight samples:

$$\hat{p}(y \mid \mathbf{x}) = \frac{1}{S}\sum_{s=1}^{S} p(y \mid \mathbf{x}, \mathbf{w}^{(s)})$$

The choice of S is arbitrary, uniform across inputs, and provides no bound on the finite-sample error of the resulting prediction. For inputs far from the decision boundary, $S = 5$ is likely sufficient; for borderline cases, $S = 1000$ may still be insufficient. A method that terminates sampling when the decision is resolved—with a provable guarantee—would improve both reliability and efficiency simultaneously.

Why Existing Approaches Fall Short

Fixed-S inference ignores input difficulty, over-sampling easy inputs and providing no guarantee on hard ones.

Heuristic early stopping (stopping when the running mean stabilises) provides no formal guarantee and empirically fails near decision boundaries.

Conformal prediction provides marginal set-coverage guarantees over the test set, not per-input anytime guarantees tied to the sampling process.

Sequential hypothesis testing (SPRT) was designed for binary tests; multi-class prediction and distributional estimation require a different treatment.

Core Mechanism

A **confidence sequence** $\{C_s\}_{s=1}^{\infty}$ satisfies:

$$\Pr\big[\exists s \geq 1 : \theta \notin C_s\big] \leq \alpha$$

This anytime-valid property allows stopping at any data-dependent time τ without inflating the Type I error. Unlike classical confidence intervals (valid only at a fixed s), confidence sequences shrink as more samples arrive and remain valid throughout.

The paper instantiates confidence sequences for three BNN tasks: (1) argmax class identification, (2) full predictive distribution estimation, and (3) probability-threshold decisions.

Technical Formulation

Let $\hat{p}_s^c = \frac{1}{s}\sum_{j=1}^{s} p(y = c \mid \mathbf{x}, \mathbf{w}^{(j)})$. A KL-mixture confidence sequence gives an anytime interval (L_s^c, U_s^c):

$$\Pr\Big[\exists s \geq 1 : p^c \notin (L_s^c, U_s^c)\Big] \leq \alpha$$

The stopping rule for argmax identification is:

$$\tau^* = \min\Big\{s : U_s^{c^*} > \max_{c \neq c^*} U_s^c\Big\}$$

where $c^* = \arg\max_c \hat{p}_s^c$.

For distributional estimation with tolerance ε:

$$\tau^{\varepsilon} = \min\Big\{s : r_s \leq \varepsilon\Big\}, \quad r_s = O\left(\sqrt{\frac{\log s}{s}}\right)$$

For threshold decisions at level τ_{thr}:

$$\tau^{\tau_{\text{thr}}} = \min\Big\{s : L_s^{c^*} > \tau_{\text{thr}} \text{ or } U_s^{c^*} < \tau_{\text{thr}}\Big\}$$

Learning or Inference Procedure

At inference time, for each input $\mathbf{x}$:

1. Draw $\mathbf{w}^{(s)} \sim q(\mathbf{w} \mid \mathcal{D})$ and evaluate $p(y \mid \mathbf{x}, \mathbf{w}^{(s)})$.
2. Update running mean $\hat{\mathbf{p}}_s$ and CS bounds (L_s^c, U_s^c).
3. Check the stopping criterion for the chosen task.
4. If met, return the prediction with its guarantee; else draw $s+1$.

No training modifications are required; the method wraps any BNN.

What the Guarantee Says

For any data-dependent stopping time τ induced by the confidence sequence and any error level α, the decision at time τ has error probability at most α—regardless of the input distribution or Monte Carlo sample realisations.

Experimental Findings

- The method assigns **more samples to ambiguous inputs** (near decision boundaries) and fewer to easy inputs, naturally concentrating compute.
- **Overall inference latency is reduced** vs. a fixed-S budget that provides the same average reliability.
- CS bounds are **tight in practice**: average sample count closely tracks input difficulty measured by entropy.
- For threshold decisions, unnecessary high-sample expenditure on clearly low- or high-uncertainty inputs is avoided.

Ablations and Interpretation

Confidence level α: Stricter guarantees (lower α) require more samples per input; the relationship is approximately $O(\log(1/\alpha))$.

Posterior quality: Better-calibrated posteriors tighten CS bounds faster; an over-dispersed posterior still provides a valid guarantee but at greater sample cost.

Heuristic early stopping: Provides no guarantee and fails empirically on high-entropy inputs where the running mean converges slowly.

Dataset difficulty skew: Savings are largest on datasets with many easy and few hard examples—the common case in practice.

Reference

Fabian Denoodt, Sibylle Hess. **When Has a Bayesian Neural Network Sampled Enough? Adaptive Inference Time with Statistical Guarantees**. NeurIPS 2026 Workshop (E-Values: From Statistics to ML). arXiv:2610.12212, October 2026.
https://arxiv.org/abs/2610.12212

Keywords: diffusion models • pixel-space generation • text-to-image • VAE-free

Iris-3B: Going Beyond the Latent with Pixel-Space Diffusion Training, Conversion and Fine-Tuning

A 3B-Parameter VAE-Free Diffusion Model Matches Latent Baselines on Text-to-Image Quality but Finds No Advantage on Fine-Grained Pixel Tasks

By Hanqiu Li Cai & Chema Garabito (Speridlabs)

arXiv:2610.09450 • October 2026

Iris-3B is a 3-billion-parameter text-to-image diffusion transformer trained entirely in pixel space—no VAE encoder, no latent compression—via a 256→512→1024 resolution curriculum, achieving GenEval 0.798 and DPG-Bench 86.52, matching latent FLUX.2 Klein. Fine-tuning on depth estimation and super-resolution finds no improvement over the latent baseline, suggesting the VAE bottleneck is less limiting than assumed.

AT A GLANCE

Problem: VAE bottleneck in latent diffusion may limit pixel-level task quality; direct pixel-space diffusion is a natural alternative.

Why it matters: If pixel-space diffusion is competitive, the field can revisit the latent paradigm assumption.

Method: PiT pixel-space diffusion transformer, 3B parameters, trained with a 256→512→1024 resolution curriculum.

Result: GenEval 0.798, DPG-Bench 86.52; no gain on depth estimation or 4× super-resolution over latent baselines.

Evidence: Compared against latent FLUX.2 Klein fine-tunes on downstream tasks; converted pixel model underperforms both.

The Big Picture

The dominant paradigm in text-to-image generation encodes images into a latent space with a VAE before denoising with a diffusion model. This reduces compute: a 1024×1024 image is processed as a ∼128×128 latent, cutting memory and FLOPs by up to 64×. But the VAE reconstruction introduces artefacts—blurring, ringing, and colour shift—that may degrade performance on pixel-level tasks such as depth estimation and super-resolution.

Iris-3B asks: does removing the VAE bottleneck and training directly in pixel space yield measurable gains on these tasks? The answer, after a thorough empirical investigation, is: not yet.

Why Existing Approaches Fall Short

Latent diffusion models (LDMs) incur VAE reconstruction error at inference time. For generation quality benchmarks, this error is tolerated; for fine-grained pixel tasks, it may be a genuine bottleneck.

Pixel-space diffusion at high resolution is expensive: the denoising network must process full-resolution feature maps, scaling quadratically in spatial dimensions.

Cascaded super-resolution (used in early pixel-space models like Imagen) adds architectural complexity and requires training separate upsampling stages.

Latent-to-pixel conversion (fine-tuning a pre-trained latent model in pixel space) is unexplored as an alternative to training from scratch.

Core Mechanism

Iris-3B (from scratch): A PiT (Pixel-space Image Transformer) diffusion model with 3B parameters. Training uses a progressive resolution curriculum: stage 1 at 256^2, stage 2 at 512^2, stage 3 at 1024^2. Text conditioning uses Qwen3-VL-4B-Instruct as the text encoder.

Converted pixel model: FLUX.2 Klein 4B is adapted to pixel space by replacing the VAE encoder/decoder with an identity mapping and adjusting channel dimensions, then fine-tuned on the pixel-space denoising objective.

Both are then fine-tuned on downstream tasks (monocular depth, 4× super-resolution).

Technical Formulation

The pixel-space denoising objective is standard denoising score matching:

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\Big[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\|_2^2\Big]$$

where $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}$ is the noisy image at timestep t and $\mathbf{c}$ is the text embedding.

The progressive training schedule uses a staged sequence $(256^2, T_1) \to (512^2, T_2) \to (1024^2, T_3)$ with decreasing step counts at higher resolutions to avoid excessive compute at the final stage.

For task-specific fine-tuning (e.g., depth estimation with ground-truth $\mathbf{d}$):

$$\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{DSM}}(\mathbf{d}) + \lambda\|\hat{\mathbf{d}} - \mathbf{d}\|_1$$

Learning or Inference Procedure

Pre-training:

1. Stage 1: train at 256^2 on image–caption pairs for T_1 steps.
2. Stage 2: upsample, continue at 512^2 for T_2 steps.
3. Stage 3: continue at 1024^2 for T_3 steps.

Fine-tuning: Replace conditioning input with the task signal; train with the task-specific loss for a small number of steps.

Inference: Standard DDPM/DDIM sampling; no VAE decoding.

What the Guarantee Says

No theoretical guarantees are claimed. The paper makes empirical claims on GenEval and DPG-Bench (generation quality) and on monocular depth and DIV2K super-resolution (fine-grained tasks).

Experimental Findings

- **Generation quality:** Iris-3B achieves **GenEval 0.798** and **DPG-Bench 86.52** at 1024^2, matching Qwen-Image and competitive with latent FLUX.2 Klein.
- **Monocular depth:** Iris-3B performs at parity with a latent FLUX.2 Klein fine-tune; the converted pixel model performs worse.
- **4× super-resolution (DIV2K):** Neither Iris-3B nor the converted model outperforms the latent FLUX.2 Klein fine-tune.
- **Inference cost:** Pixel-space at 1024^2 is substantially more expensive per step than latent inference.

Ablations and Interpretation

Curriculum stages: Skipping the 256^2 stage degrades final generation quality; low-frequency pre-training is important for convergence.

Conversion quality: The converted pixel model underperforms both the from-scratch model and the original latent model, suggesting that the latent representation carries useful inductive bias.

Text encoder: Replacing Qwen3-VL-4B with a smaller encoder degrades text alignment (DPG) more than object-count metrics (GenEval).

Interpretation: The null result suggests that either the VAE bottleneck is less harmful than assumed, that the latent representation is beneficial beyond compression, or that pixel-space models need more task-specific data to close the gap.

Reference

Hanqiu Li Cai, Chema Garabito (Speridlabs). **Iris-3B: Going Beyond the Latent with Pixel-Space Diffusion Training, Conversion and Fine-Tuning**. arXiv:2610.09450, October 2026.
https://arxiv.org/abs/2610.09450